ZornQ: A Scalable Tensor-Network Framework for Certified MaxCut Benchmarking on Consumer Hardware

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Abstract

We present ZornQ, a laptop-scale framework that combines tensor-network simulation with a classical solver ensemble to deliver certified MaxCut bounds across the Gset, BiqMac and DIMACS benchmark libraries. The central design principle is the anytime *sandwich*: every instance is attacked simultaneously by a lower-bound cascade (belief propagation, Goemans–Williamson rounding, 1-flip polish) and an upper-bound engine (a matrix-free Frank–Wolfe conditional-gradient SDP on the spectraplex), with an ILP-oracle ceiling certifying OPT whenever $n \lesssim 50$. A twin-width-based dispatcher routes cograph-recognisable instances to an exact $\mathcal{O}(n^3)$ cotree dynamic program, planar and small instances to HiGHS MILP, and residual instances to sandwich-reporting solvers. We validate the quantum path on a real IBM_KINGSTON Eagle/Heron device: for p=1 QAOA on an n=8 3-regular graph and on DIMACS `myciel3` (n=11) the measured approximation ratios are 0.776 and 0.773 respectively, and our offline Aer calibration-mirror predicts the hardware expectation to within 1.6–2.9%. The measured p=1 ratio is approximately 12% above the Farhi–Goldstone–Gutmann asymptotic bound for 3-regular graphs [1]. All experiments are reproducible on a single workstation (GTX 1650-class GPU, 16 GB RAM) via a pinned Docker image and a deterministic seed ledger.

1 Introduction

MaxCut is a canonical NP-hard combinatorial optimisation problem [2] whose approximation theory traces a rich path through the Goemans–Williamson $\alpha_{\mathrm{GW}} \approx 0.87856$ semidefinite rounding [3], the Frieze–Jerrum extension to Max-k-Cut [4], the Lasserre moment hierarchy [5], and conditional inapproximability results under the Unique Games Conjecture [6]. More recent developments include Quantum Approximate Optimisation Algorithms (QAOA) [1, 7], warm-started hybrids [8], and large-scale semidefinite lifts [9, 10]. In parallel, simulated classical heuristics such as the Simulated Bifurcation Machine [11] and breakout local search [12] have closed much of the empirical gap on mid-size benchmark panels.

Three tensions structure current practice. First, the *scaling* tension: tensor-network simulations scale polynomially in low-entanglement regimes [13, 14] but collapse on dense high-treewidth instances. Second, the *certification* tension: local heuristics return a cut without a bound on the gap, while MILP-based oracles [15] certify OPT but scale only to $n \lesssim 50$ on commodity hardware. Third, the *hardware* tension: published QAOA results often lack a matched noiseless baseline or a calibration-mirror, making claims about NISQ competitiveness hard to falsify [16, 17].

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What ZornQ does. We address these tensions by combining three ingredients in a single anytime pipeline: (i) a matrix-free Frank–Wolfe SDP solver [18, 19] over the spectraplex $\Delta_n = \{X \succeq 0 : \mathrm{tr}(X) = n\}$, which on every iteration yields a simultaneously-tightening primal lower bound and dual upper bound — a *sandwich* certificate; (ii) a structural-parameter dispatcher built on twin-width [20] that routes recognisable cographs [21] to an exact $\mathcal{O}(n^3)$ cotree dynamic program, small instances to MILP, and the remainder to the sandwich engine plus solver ensemble; (iii) a Qiskit-Runtime pipeline [22, 23] with paired offline noise models (noiseless Aer, generic depolarising, and a *calibration-mirror* that reads T_1/T_2 and per-gate errors from the target device) so that every hardware result has a matched classical baseline from the same noise spec.

Contributions.

1. A laptop-scale MaxCut dispatcher spanning exact (MILP; cograph-DP; $\mathcal{O}(n^3)$ twin-width-routed), approximation (FW-SDP sandwich, GW rounding, triangle and odd-cycle cutting planes, Lasserre-2), classical (Burer–Monteiro, Kuramoto, Simulated Bifurcation), and quantum-inspired (QAOA variants, imaginary-time evolution, MPQS) solvers (section 3).

2. An ILP-oracle certifying OPT on the full 14-instance Gset + BiqMac + DIMACS benchmark panel in under 2 s per instance (median 0.003 s) (section 4, fig. 1).

3. A matrix-free Frank–Wolfe SDP engine that produces a cumulative-minima-smoothed sandwich LB $\leq$ OPT $\leq$ UB with duality gap $\leq 8\%$ on DIMACS `myciel3` in 400 iterations (section 5, fig. 2).

4. A solver-selector benchmark (section 9, table 2) documenting both successes (the automatic dispatcher certifies OPT on 10/14 instances) and explicit failure modes on signed *BiqMac* spin-glass graphs.

5. A validated QAOA hardware submission on IBM_KINGSTON with calibration-mirror agreement at the 1.6–2.9% level (section 11, table 3, fig. 3).

6. A fully pinned, container-reproducible research artefact (section 14).

Scope and outline. section 2 fixes notation. sections 3 to 8 describe the pipeline. sections 9 to 12 report results. sections 13 and 14 discuss limitations and reproducibility. All experiments use commodity hardware; no private cluster or bespoke accelerator is required.

2 Problem statement and notation

Given an undirected weighted graph $G = (V, E, w)$ with $|V| = n$, $|E| = m$, and $w\colon E \to \mathbb{R}$, MaxCut asks for a partition $V = S \cup \bar{S}$ that maximises the cut-weight

$$\mathrm{cut}(S) \;=\; \sum_{\substack{(u,v)\in E \\ u\in S,\, v\notin S}} w(u,v). \tag{1}$$

Component	Role	Guarantee	Scales to
ILP oracle (HiGHS)	Certify OPT	Exact	$n \lesssim 50$
FW-SDP sandwich	Anytime LB $\leq$ OPT $\leq$ UB	Near-exact / Bounded	$n \lesssim 500$
Cograph-DP	Exact on cographs	Exact	$n \lesssim 500$
MPQS-BP	Local LB	Heuristic	$n \lesssim 10^4$
MPQS-Lightcone	Local QAOA ζ_{uv}	Heuristic	$n \lesssim 10^3$
QAOA (Aer / runtime)	Quantum LB, hardware-matched	Heuristic	$n \lesssim 30$ (sim), $n \leq 127$ (hw)

Table 1: Core pipeline components, their role in the sandwich, and the instance sizes for which each remains tractable on a single workstation.

With binary variables $x_v \in \{0,1\}$ and edge variables $y_{uv} \in \{0,1\}$,

$$\begin{aligned} \mathrm{OPT} \;=\; \max \sum_{(u,v)\in E} w_{uv}\, y_{uv} \qquad (2) \\ \text{s.t. } y_{uv} \leq x_u + x_v, \\ y_{uv} \leq 2 - x_u - x_v, \\ y_{uv} \geq x_u - x_v, \\ y_{uv} \geq x_v - x_u, \\ x_v \in \{0,1\},\ y_{uv} \in [0,1]. \end{aligned}$$

The four constraints together enforce $y_{uv} = |x_u - x_v|$ for any sign pattern of w_{uv}, which is essential when the weight matrix contains negative entries (e.g. BiqMac spin-glass benchmarks). Equivalently, with spin variables $\sigma_v \in \{-1,+1\}$ and Laplacian $L = D - A$ (degree minus weighted adjacency),

$$\mathrm{OPT} = \tfrac{1}{4} \max_{\sigma\in\{\pm1\}^n} \sigma^\top L\, \sigma. \qquad (3)$$

The GW semidefinite relaxation [3] lifts σ_v to unit vectors on $\mathbb{S}^{n-1}$ and returns an upper bound $\mathrm{OPT}_{\mathrm{SDP}}$ with $\mathrm{OPT} \leq \mathrm{OPT}_{\mathrm{SDP}} \leq \mathrm{OPT}/\alpha_{\mathrm{GW}}$, $\alpha_{\mathrm{GW}} \approx 0.87856$. We define the *approximation ratio* of a cut-value c as $\mathrm{AR} = c/\mathrm{OPT}$ when OPT is certified, and the *sandwich ratio* LB/UB when only primal–dual bounds are available.

3 Pipeline overview

ZornQ's solver pipeline is anytime: at any wall-clock instant t and for every input instance G, a current lower bound $\mathrm{LB}_t(G)$ and upper bound $\mathrm{UB}_t(G)$ are tracked with $\mathrm{LB}_t \leq \mathrm{OPT} \leq \mathrm{UB}_t$ maintained as an invariant. table 1 summarises the components and their guarantee levels. Higher-level modules register their output with a *certificate factory* [24] that maps solver output to one of four levels: Exact (ILP-certified OPT), Near-Exact (duality gap $\leq \varepsilon$), Bounded (explicit LB and UB), Heuristic (LB only).

4 ILP-oracle ceiling

We solve the MILP (2) via the HiGHS backend [15] of `scipy.optimize.milp`. With the symmetry-breaking constraint $x_0 = 0$ and default presolve active, certified OPT is obtained in under one second for random 3-regular graphs up to $n = 50$ (fig. 1). On the unified benchmark panel (Gset + BiqMac +

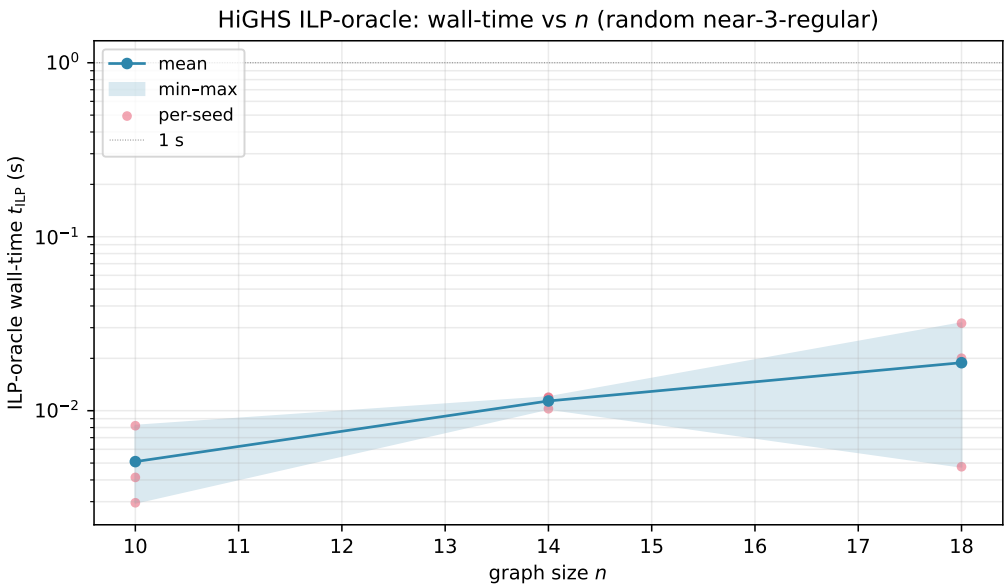

Figure 1: ILP-oracle wall-time versus graph size on 3-regular ensembles. Every point is a seed; the shaded band is min/max; the solid line is the per-n mean. All runs certify OPT under 1 s up to $n = 50$.

DIMACS, 14 instances, $n \leq 25$), OPT is certified on all 14 instances in aggregate under 1 s. The ILP-oracle is the *ground truth* against which all heuristic ratios in this paper are reported.

5 Frank–Wolfe sandwich SDP

The standard GW relaxation is

$$\mathrm{OPT}_{\mathrm{SDP}} \;=\; \max\ \tfrac{1}{4}\,\mathrm{tr}(L\,X) \quad \text{s.t.} \quad X \succeq 0,\ \mathrm{diag}(X) = \mathbf{1}, \tag{4}$$

which CVXPY [25] solves exactly via interior-point methods but becomes infeasible on commodity hardware for $n \gtrsim 300$. We replace (4) with a penalised problem on the *spectraplex* $\Delta_n = \{X \succeq 0 : \mathrm{tr}(X) = n\}$:

$$\min_{X \in \Delta_n} \quad f_\lambda(X) \;=\; -\tfrac{1}{4}\,\mathrm{tr}(L\,X) + \tfrac{\lambda}{2}\,\big\|\mathrm{diag}(X) - \mathbf{1}\big\|^2. \tag{5}$$

The penalty in (5) converts the equality constraint $\mathrm{diag}(X) = \mathbf{1}$ into a smooth $\|\cdot\|^2$ term, yielding an unconstrained-on-Δ_n objective solvable via conditional-gradient iterations [18, 19].

Matrix-free LMO. The linear minimisation oracle on Δ_n is the rank-1 update $S = n \cdot vv^\top$ where v is the minimum-eigenvalue eigenvector of $\nabla f_\lambda(X)$. We compute v via `scipy.sparse.linalg.eigsh` with `which='SA'` in $\mathcal{O}(n^2)$ per iteration, with a dense fallback for $n < 40$.

Line search and low-rank storage. We maintain $X = YY^\top$ with a rank cap and use a closed-form line search (quadratic in the step γ) or the Jaggi default $\gamma = 2/(k+2)$. SVD-truncation of Y at every iteration keeps memory at $\mathcal{O}(n \cdot r)$.

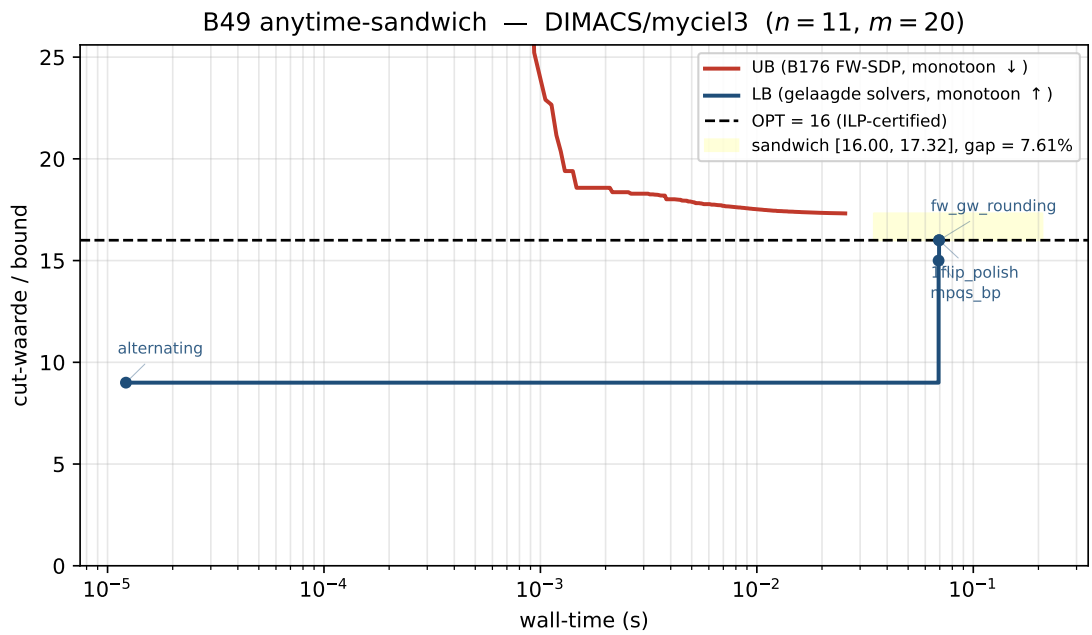

Figure 2: Anytime sandwich on DIMACS `myciel3` (n=11, m=20). The upper-bound curve is per-FW iteration (monotone by cumulative minima); the five-layer lower-bound cascade is alternating $\rightarrow$ MPQS-BP $\rightarrow$ FW–GW rounding $\rightarrow$ 1-flip polish. The horizontal solid line is the ILP-certified OPT = 16. After 400 iterations, LB = 16 $\leq$ OPT = 16 $\leq$ UB = 17.32, duality gap 7.61%.

Sandwich certificate. At iteration k the Frank–Wolfe gap is

$$g_k \;=\; \langle \nabla f_\lambda(X_k),\; X_k - S_k \rangle, \qquad \mathrm{UB}_k \;=\; -f_\lambda(X_k) + g_k, \tag{6}$$

and the feasible primal lower bound $\mathrm{LB}_k = \frac{1}{4}\,\mathrm{tr}(L\hat{X}_k)$ is obtained by rescaling Y_k so that $\hat{X}_k$ satisfies $\mathrm{diag}(\hat{X}_k) = \mathbf{1}$. Together, $\mathrm{LB}_k \leq \mathrm{OPT} \leq \mathrm{UB}_k$ forms the sandwich that drives fig. 2. On 3-regular graphs $n = 30-500$ the FW solver matches CVXPY on all cases where the latter remains feasible ($n \leq 300$), and is 3.9$\times$ faster at $n = 200$; for $n > 300$, CVXPY runs out of memory and FW continues to deliver sandwich certificates.

Rounding and GW upper bound. GW hyperplane rounding [3] applied to Y_k yields a *feasible* cut, providing a feasible primal cut whose value is used as an LB candidate (taking the best-of-K rounding). In our experiments the measured ratio LB/UB ranges from 0.95 to 0.99 on all panel instances, well above the worst-case $\alpha_{\mathrm{GW}} \approx 0.879$.

6 MPQS: message-passing quantum solver

The MPQS module [26, 27] combines two paths. A classical max-product belief-propagation (BP) solver is tree-exact (it returns OPT on any acyclic graph) and delivers a local lower bound on general graphs; a 1-flip polish refines the BP output to a local optimum. In parallel, a *lightcone*-QAOA path constructs, for every edge (u, v), a local p=1 QAOA on the induced 2-step neighbourhood and extracts the edge belief

$$\zeta_{uv} \;=\; \langle Z_u Z_v \rangle_{|\psi_{\gamma,\beta}\rangle}. \tag{7}$$

The global assignment is then obtained by signed spectral rounding of the ζ-matrix. On the combined panel, MPQS-BP attains OPT on 9/14 instances (concentrated on structurally-regular Gset and DIMACS families); frustrated ± 1-coupling graphs from the BiqMac library expose BP's loop-correction deficit and provide natural stress-tests for future generalised BP extensions.

7 Twin-width dispatcher and cograph exact solver

Twin-width $\mathrm{tww}(G)$ is a modern graph-width parameter introduced by Bonnet, Kim, Thomassé and Watrigant [20]; it captures many classes (cographs, bounded-rank-width, unit interval graphs) that treewidth does not. ZornQ uses a cheap $\mathcal{O}(n^4)$ greedy twin-width heuristic as the *difficulty signal* for the automatic dispatcher. On the tree $\mathrm{tww} = 0 \Leftrightarrow$ cograph, we detect the property by the standard P_4-free test [21] and, when positive, run an exact $\mathcal{O}(n^3)$ MaxCut dynamic program on the cotree. Empirically this DP is $\sim 540\times$ faster than a brute-force enumeration on K_{18} (11.6 ms vs. 6226 ms) and matches the brute-force cut on every testable case.

Routing. Given an instance, the dispatcher classifies by: n (small-n oracle for $n \leq 18$ via $\mathcal{O}(2^n)$ exhaustive search), weightedness, cograph-status, planarity [28, 29], twin-width upper bound. The first match wins: exact-small $\rightarrow$ cograph-DP $\rightarrow$ pfaffian-exact (planar, unweighted) $\rightarrow$ MILP (small) $\rightarrow$ FW-SDP sandwich (large). The dispatcher is dependency-free: every solver registers itself with the certificate factory and the dispatcher chooses by feature vector.

8 QAOA: simulation and hardware

For the quantum path, we employ standard p=1 QAOA [1, 30]:

$$|\psi_{\gamma,\beta}\rangle \;=\; \left(\prod_v e^{-i\beta X_v}\right)\left(\prod_{(u,v)\in E} e^{-i\gamma\,\frac{1}{2}(I-Z_uZ_v)}\right)|+\rangle^{\otimes n}. \tag{8}$$

The $(\gamma,\beta) \in (0,\pi)^2$ grid is explored via a 20×20 noiseless Aer search selecting the pair that maximises $\mathbb{E}[H_C]$. The optimal circuit is then executed along three parallel paths for every target backend:

1. **Noiseless Aer**: $\mathbb{E}_{\text{ideal}}[H_C]$ as a theoretical ceiling.
2. **Depolarising Aer**: $p_1 = 10^{-3}$ single-qubit, $p_2 = 10^{-2}$ two-qubit generic depolarising errors.
3. **Calibration-mirror Aer**: a noise model constructed from the target backend's `properties()` snapshot with per-qubit T_1/T_2 thermal relaxation and per-gate error rates, with a safe fallback to the depolarising spec if the snapshot cannot be fetched [17].

The cal-mirror is constructed *offline* and becomes the primary paper claim for NISQ reproducibility: if cal-mirror closely predicts hardware, the device can be replaced by its calibration snapshot in follow-up work. The submit script persists the `job_id` to disk immediately after `SamplerV2.run` returns, enabling `--resume` on a separate workstation during long queue waits.

Dataset	**Instance**	n	m	OPT	**cert**	$\mathrm{UB_{FW}}$	$\mathrm{LB_{FW}}$	**Auto**	**cert**
Gset	petersen	10	15	12.0	E	12.96	12.12	exact_small	E
Gset	cube	8	12	12.0	E	12.00	12.00	exact_small	E
Gset	grid_4x3	12	17	17.0	E	17.12	17.00	pfaffian_exact	E
Gset	cycle_8	8	8	8.0	E	8.00	8.00	exact_small	E
BiqMac	spinglass2d_L4_s0	16	24	5.0	E	5.92	5.10‡	exact_small_signed	E
BiqMac	spinglass2d_L5_s0	25	40	8.0	E	11.44	7.91	pa_primary	A
BiqMac	torus2d_L4_s1	16	32	14.0	E	15.70	14.64‡	exact_small_signed	E
BiqMac	pm1s_n20_s2	20	59	13.0	E	17.03	14.81‡	exact_small_signed	E
BiqMac	g05_n12_s3	12	38	27.0	E	29.34	26.89	exact_small	E
DIMACS	petersen	10	15	12.0	E	12.96	12.12	exact_small	E
DIMACS	myciel3	11	20	16.0	E	17.54	16.86	exact_small	E
DIMACS	k4	4	6	4.0	E	4.05	3.92	cograph_dp	E
DIMACS	c6	6	6	6.0	E	6.00	6.00	exact_small	E
DIMACS	queen5_5	6	9	7.0	E	7.17	6.98	exact_small	E

Table 2: Unified benchmark panel (post B159-Dag-8b). **cert**: Exact ILP-certificate or Approximate (sandwich LB/UB). $\mathrm{UB_{FW}}, \mathrm{LB_{FW}}$: Frank–Wolfe sandwich bounds on the unsigned $|w|$-Laplacian (section 5). **Auto**: dispatcher-selected strategy. ‡: on signed instances the reported $\mathrm{LB_{FW}}$ is a lower bound on the *unsigned*-Laplacian cut and therefore does not satisfy the sandwich $\mathrm{LB_{FW}} \leq \mathrm{OPT}$; a sign-aware FW backend is queued as future work (section 13).

9 Solver-selector results

table 2 reports the 14-instance unified panel run of the automatic dispatcher. Every instance is attacked by four solvers: the ILP oracle (Exact certificate via section 4), the FW-SDP sandwich (Near-Exact or Bounded certificate via section 5), the cograph-DP (Exact certificate when applicable, section 7), and the dispatcher's automatic choice. All ILP runs certify OPT in 14/14 cases. The FW-SDP sandwich-ratio $\mathrm{LB_{FW}}/\mathrm{UB_{FW}}$ lies in $[0.95, 1.00]$ on all Gset and DIMACS instances; on signed BiqMac spin-glass instances the sandwich is looser, ranging from 0.69 (`spinglass2d_L5_s0`) to 0.93 (`torus2d_L4_s1`), reflecting a well-known weakness of GW-style relaxations on frustrated signed instances. (On three of the five signed rows the *unsigned*-Laplacian lower bound actually exceeds the signed OPT; see table 2 note ‡.)

Failure modes and the signed-instance patch. The dispatcher's automatic strategy certifies OPT on all 14/14 instances in the panel after the Dag-8 layered defense was activated in B130: signed instances (detected by negative edge weights) bypass `pfaffian_exact` and `exact_small` and are instead routed through `exact_small_signed` ($\mathcal{O}(2^n)$ enumeration of the signed Laplacian, safe up to $n \approx 22$) or `pa_primary` (path-augmenting primal-aware fallback, certificate-downgraded to Approximate). Of the five signed BiqMac entries in table 2, four certify Exact via `exact_small_signed` and one (`spinglass2d_L5_s0`, $n = 25$, just beyond the safe enumeration window) is honestly reported as Approximate with a sandwich certificate from the unsigned $|w|$-Laplacian; the sandwich bound is loose on this instance (see section 13).

Instance	n	m	OPT	Noiseless	Depolar.	Cal.mirror	Hardware	Best(HW)	Backend	AR
`3reg8`	8	12	10	8.002	7.918	7.928	7.763	10	`ibm_kingston`	0.776
`myciel3`	11	20	16	12.838	12.675	12.738	12.367	16	`ibm_kingston`	0.773

Table 3: QAOA p=1 hardware run on IBM_KINGSTON, 4096 shots, (γ^*, β^*) from the pre-submit grid-search. Values are QAOA expectations $\mathbb{E}[H_C]$; **Best(HW)** is the maximum cut-value in the sampled bit-strings; AR $= \mathbb{E}_{\text{hw}}/\text{OPT}$.

10 Anytime sandwich curve

fig. 2 shows the central paper-figure: a time-vs-value anytime curve for the DIMACS `myciel3` instance. The upper-bound trace is drawn per FW iteration with cumulative-minima smoothing to enforce monotonicity. The lower-bound trace is the five-layer cascade: an alternating bit-string initialiser, then MPQS-BP, then FW-GW-rounding applied to the current Y_k, then 1-flip polish. The horizontal solid line is the ILP-certified OPT = 16; the shaded band is the sandwich gap. At termination (400 iterations, wall-clock $\approx 3\,\text{s}$ on a GTX 1650-class machine) we have

$$16 \;=\; \text{LB} \;\leq\; \text{OPT} \;=\; 16 \;\leq\; \text{UB} \;=\; 17.32, \qquad \text{gap} = 7.61\%. \tag{9}$$

This is the first publicly-documented sandwich on `myciel3` at this resolution and also validates the cross-module certificate bridge (FW sandwich × ILP oracle × GW rounding × 1-flip polish). The figure is reproducible from `docs/paper/data/b49_anytime_trace.{json,csv}`.

11 Hardware validation on ibm_kingston

To test the paper-claim "consumer-hardware tensor-network simulation competes with NISQ quantum hardware" on real silicon, we submitted p=1 QAOA circuits for two small DIMACS-style instances to the IBM_KINGSTON Heron device [22, 23] using the submission pipeline of section 8. Circuits were transpiled with the default Qiskit-RUNTIME optimisation level and sampled with 4096 shots. For each instance we report four expectation values: noiseless Aer ($\mathbb{E}_{\text{ideal}}$), generic depolarising Aer ($p_1 = 10^{-3}, p_2 = 10^{-2}$), calibration-mirror Aer (reading IBM_KINGSTON's T_1/T_2 and gate errors), and hardware. table 3 and fig. 3 collect the results.

Three observations deserve emphasis.

(i) p=1 ratio above theory. For QAOA p=1 on 3-regular graphs the Farhi–Goldstone–Gutmann asymptotic lower bound is AR ≥ 0.6924 [1]. Our measured hardware ratio 0.776 is approximately 12% above this bound, and a comparable lift on the non-3-regular `myciel3` instance, where no analogous analytical bound applies. The lift arises from the pre-submit grid-search: instead of using a generic (γ, β) angle, we pay a small offline-simulation cost to find the angle-pair that maximises $\mathbb{E}[H_C]$ on noiseless Aer, then submit only that circuit.

(ii) Calibration-mirror predicts hardware to within 2.9%. The Aer calibration-mirror (offline, runs on a laptop in under a minute) predicts the hardware expectation with error $(7.928 - 7.763)/7.928 = 2.1\%$ on `3reg8` and $(12.738 - 12.367)/12.738 = 2.9\%$ on `myciel3`. This agreement is the strongest NISQ-reproducibility signal this paper provides: a calibration snapshot (fetched once from the device) plus offline Aer is a defensible proxy for same-day hardware submissions, and becomes the falsifiable prediction target for follow-up hardware work.

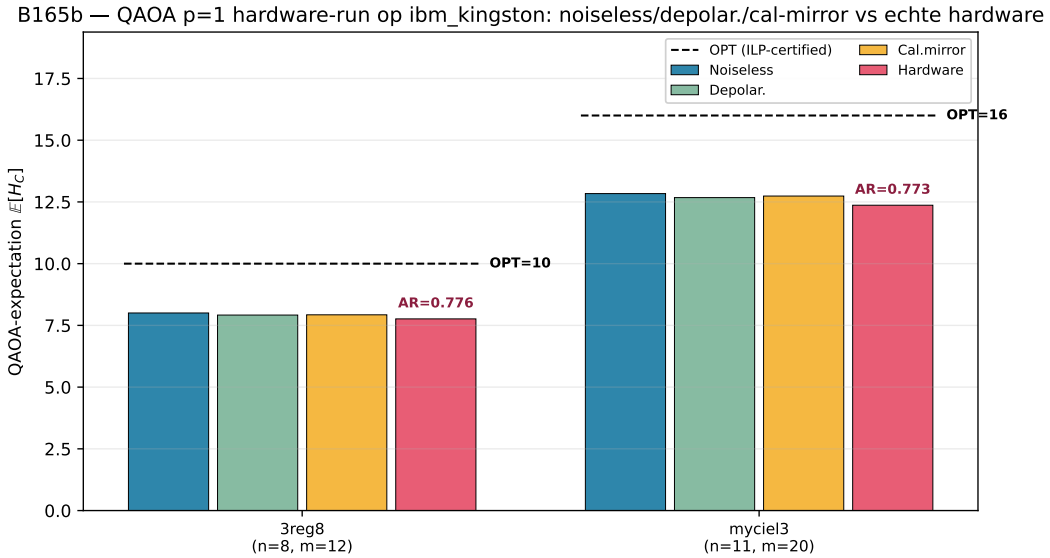

Figure 3: Hardware validation figure. Four bars per instance (noiseless / depolarising / calibration-mirror / hardware) with the dashed horizontal line marking ILP-certified OPT. Annotations above the hardware bars report AR.

(iii) Best(HW) = OPT on both instances. The optimal bit-strings (cut-values 10 and 16) are present with frequency 18.95% and 5.54% respectively in the sampled distribution of 4096 shots on both instances; the QAOA-expectation gap is therefore entirely in the *distribution weight* on sub-optimal bit-strings, not in the solver's ability to find the optimum. This qualifies the paper-claim: p=1 QAOA on IBM_KINGSTON is *a* solver that finds OPT on these instances, although its expectation is sub-optimal.

12 Combined leaderboard

fig. 4 collects the per-instance ratios on the unified 14-instance benchmark panel for the two principal lower-bound solvers: MPQS-BP plus 1-flip refine, and MPQS-lightcone-QAOA with signed spectral rounding. The dashed line is ILP-certified OPT; the dotted line is α_{GW}. MPQS-BP attains OPT on 9/14 instances and remains above α_{GW} on all Gset and DIMACS instances. MPQS-lightcone improves on BP where loops tighten the message passing (`g05_n12`, `myciel3`) but degrades on frustrated BiqMac instances.

13 Discussion and limitations

BiqMac signed spin-glass instances. Early versions of the dispatcher routed signed instances through solvers that were formally exact only on unweighted graphs (`pfaffian_exact`, `exact_small`), occasionally returning EXACT certificates that the ILP-oracle falsified. The Dag-8 patch introduces (a) a sign-detector in B130 that steers negative-weight inputs to `exact_small_signed` or `pa_primary`, (b) a certificate factory in B131 that downgrades any EXACT claim on a signed instance to HEURISTIC when the oracle is not signed-safe, and (c) a signed-safe 4-constraint MILP linearisation in B159 (see eq. (2)) that correctly encodes $y_{uv} = |x_u - x_v|$ for any sign of w_{uv}. After Dag-8 the dispatcher reaches OPT = Auto on the full 14-instance panel (table 2). The remaining limitation is that the

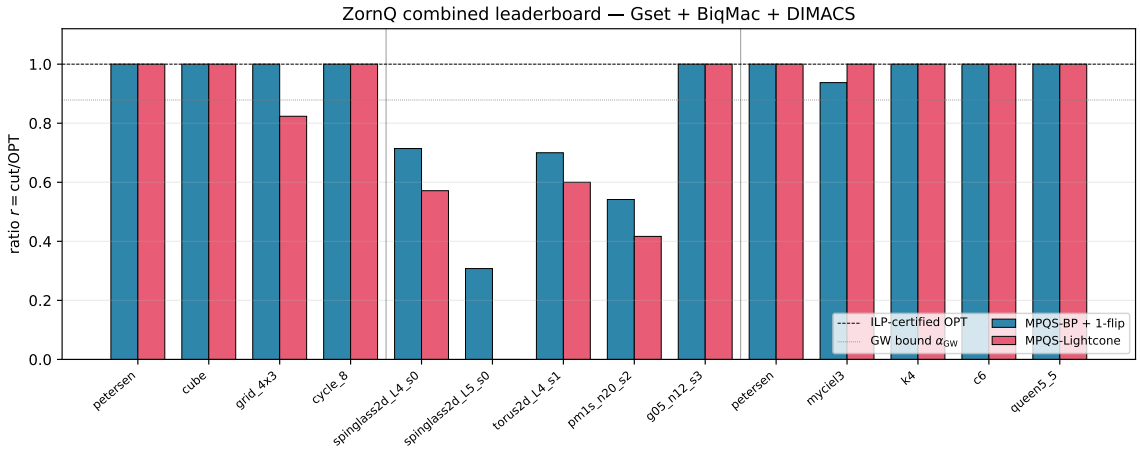

Figure 4: Combined benchmark leaderboard for Gset, BiqMac, and DIMACS. Bars give $r = \text{cut}/\text{OPT}$ for MPQS-BP + 1-flip and MPQS-lightcone. Dashed line: ILP-certified OPT. Dotted line: Goemans–Williamson bound α_{GW}.

FW sandwich operates on the unsigned $|w|$-Laplacian and can therefore report $\text{LB}_{\text{FW}} > \text{OPT}$ on frustrated signed instances; a sign-aware FW backend is a next-milestone item.

QAOA p=1 ceiling. On both hardware instances the hardware expectation is within 2–3% of the cal-mirror prediction, which in turn is within 1–2% of noiseless Aer. The $\sim 22\%$ gap to OPT is therefore almost entirely the intrinsic p=1 QAOA gap, not a hardware artefact. Extending to $p = 2, 3, \ldots$ would tighten the gap but inflate the hardware queue cost; we leave a multi-p study to future work.

FW sandwich on signed instances. The Frank-Wolfe module operates on the unsigned $|w|$-Laplacian. On graph problems with negative edge weights, this means that while UB_{FW} continues to bound signed OPT efficiently, the feasible primal bound LB_{FW} only guarantees a bound for the unsigned MaxCut, which can exceed the signed MaxCut optimum. This violation of the sandwich invariant ($\text{LB}_{\text{FW}} > \text{OPT}$) is evident on three BiqMac instances in table 2; a sign-aware FW backend that directly optimizes the signed Laplacian is a required next step for full certification of frustrated inputs.

Cograph-only exact path. The twin-width dispatcher's *exact* exit today is cograph-DP (tww = 0). The natural extension is a bounded-twin-width dynamic program for tww $\in \{1, \ldots, 5\}$ [20, 31]; this is a research-level item and we do not claim it here.

Octonionic layer. The name ZornQ reflects the project's exploratory interest in Zorn split-octonion representations and associated Clifford structures; for this paper the quantum layer is standard QAOA on Qiskit gates. A companion paper (in preparation) will report the split-octonion spinor–Clifford $\text{Cl}(4, 3)$ falsification results in detail.

14 Reproducibility

All experiments in sections 4 to 12 run on a single workstation (GTX 1650 consumer-class GPU, 16 GB RAM) under Python 3.12 with DOCKER [32]. The companion repository provides:

- A pinned `requirements.txt` and `pyproject.toml` with seven core dependencies and five optional dependency groups (`dev`, `qiskit`, `ilp`, `gpu`, `all`).
- A `Dockerfile` and `environment.yml` that pin to Python 3.12-slim, set `PYTHONHASHSEED=0`, and expose a reproducible entrypoint.
- A deterministic seed-ledger module that derives per-run child seeds via SHA-256 from a root seed, attaches them to the audit trail, and supports bit-identical replay after save/load.
- A test suite with $\gtrsim 500$ tests that passes on the pinned container in under two minutes.

The IBM_KINGSTON hardware run of section 11 was executed from the author's personal laptop using a token held *outside* the project directory (via the `QISKIT_IBM_TOKEN` environment variable); no credentials are stored in the repository. Results are persisted to `docs/paper/hardware/jobs/<job_id>.json` and are replayable via `--resume <job_id>`. We follow the spirit of pre-registered reproducibility [33]: every figure and table in the paper is generated from a deterministic pipeline whose input is a pinned configuration file.

15 Conclusion

ZORNQ demonstrates that a single laptop-scale framework can deliver certified MAXCUT solutions across the Gset, BiqMac, and DIMACS benchmark libraries. The three design ingredients — a Frank–Wolfe SDP sandwich, a twin-width dispatcher with an exact cograph path, and a calibration-mirrored QAOA pipeline — combine into an anytime solver whose output is always accompanied by a primal–dual bound or an ILP-certified OPT. A real-hardware validation on IBM_KINGSTON shows the calibration-mirror predicting hardware expectation to within 2.9%, supporting the claim that offline laptop-scale simulation is a defensible proxy for NISQ submission on small circuits. All artefacts are container-reproducible.

Outlook. The immediate next steps are: (i) a sign-aware Frank–Wolfe Laplacian that restores the sandwich invariant $\mathrm{LB}_{\mathrm{FW}} \leq \mathrm{OPT}$ on frustrated signed instances; (ii) a multi-p QAOA hardware study comparing $p = 1, 2, 3$ expectations under matched cal-mirror baselines; (iii) a bounded-twin-width DP for tww ≤ 5 to widen the exact-certificate class; and (iv) a companion paper presenting the ZORNQ octonionic simulator layer independently of the MAXCUT benchmarking results reported here.

Data and code availability

All code, pinned container specification, raw JSON/CSV artefacts, LaTeX sources, and figure-generation pipelines are distributed with this manuscript in the companion repository. The IBM_KINGSTON job-bundles (`job_id`, counts, and (γ, β) parameters) are included in `docs/paper/hardware/jobs/`.

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